

SHETH L.U.J & SIR M.V COLLEGE OF SCIENCE

SUBJECT : AI

Aim: K-Nearest Neighbors (K-NN)

Implement the K-NN algorithm for classification or regression.

Apply the K-NN algorithm to a given dataset and predict the class or value for test data.

Evaluate the accuracy or error of the predictions and analyze the results.

Input

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PRACTICAL 8.py - C:/Users/DELL/Downloads/PRACTICAL 8.py (3.13.11)
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import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

# Load dataset
df = pd.read_csv("Crop_recommendation.csv")

print("Dataset shape:", df.shape)

print("\nColumns:")
print(df.columns)

# Input features
X = df[["N", "P", "K", "temperature", "humidity", "ph", "rainfall"]]

# Target
y = df["label"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

print("\nTraining records:", len(X_train))
print("Testing records:", len(X_test))

# Feature scaling
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

# Test different K values
k_values = [1, 3, 5, 7, 9, 11]
accuracies = []

for k in k_values:
    model = KNeighborsClassifier(n_neighbors=k)
    model.fit(X_train, y_train)
    y_pred = model.predict(X_test)
    accuracy = accuracy_score(y_test, y_pred)
    accuracies.append(accuracy * 100)

# Display accuracy for each K
print("\nAccuracy for different K values:")
for k, accuracy in zip(k_values, accuracies):
    print("K =", k, "Accuracy =", round(accuracy, 2), "%")

# Best K
best_index = accuracies.index(max(accuracies))
best_k = k_values[best_index]

print("\nBest K value:", best_k)
print("Best Accuracy:", round(accuracies[best_index], 2), "%")

# Train final KNN model
model = KNeighborsClassifier(n_neighbors=best_k)
```

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# Train final KNN model
model = KNeighborsClassifier(n_neighbors=best_k)

model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Final accuracy
final_accuracy = accuracy_score(y_test, y_pred)

print("\nFinal K-NN Accuracy:")
round(final_accuracy * 100, 2), "%")

# Predict first 5 test samples
print("\nPredictions for first 5 test samples:")

for i in range(5):
    print(
        "Actual:", y_test.iloc[i],
        "Predicted:", y_pred[i]
    )

# Accuracy vs K graph
plt.figure(figsize=(7, 5))

plt.plot(k_values, accuracies, marker="o")

plt.title("K-NN Accuracy vs K")
plt.xlabel("K Value")
plt.ylabel("Accuracy (%)")

plt.xticks(k_values)
plt.grid(True)

plt.show()
```

Output

```
*IDLE Shell 3.13.11*
File Edit Shell Debug Options Window Help
Python 3.13.11 (tags/v3.13.11:6278944, Dec 5 2025, 16:26:58) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/DELL/Downloads/PRACTICAL 8.py =====
Dataset shape: (2200, 8)

Columns:
Index(['N', 'P', 'K', 'temperature', 'humidity', 'ph', 'rainfall', 'label'], dtype=object)

Training records: 1760
Testing records: 440

Accuracy for different K values:
K = 1 Accuracy = 97.95 %
K = 3 Accuracy = 97.73 %
K = 5 Accuracy = 97.95 %
K = 7 Accuracy = 97.05 %
K = 9 Accuracy = 97.73 %
K = 11 Accuracy = 97.27 %

Best K value: 1
Best Accuracy: 97.95 %

Final K-NN Accuracy: 97.95 %

Predictions for first 5 test samples:
Actual: orange Predicted: orange
Actual: banana Predicted: banana
Actual: cotton Predicted: cotton
Actual: maize Predicted: maize
Actual: orange Predicted: orange
```

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Figure 1

